

DIPLOMA THESIS

Documentation

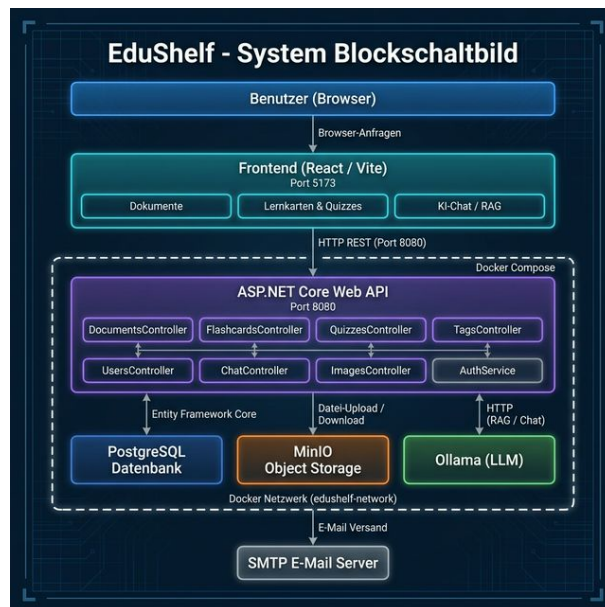
Author(s)	Elias Welt, Phillip Rubak, ,
From Academic year	2025/26
Topic	EduShelf
Co-operation partners	X-Works

Assignment of tasks	This diploma project develops 'EduShelf', a web-based platform for managing educational content. It provides students and teachers with a centralized solution for organizing documents and leveraging AI-powered information retrieval (RAG – Retrieval Augmented Generation). The system integrates a modern web frontend, a backend for data processing and storage, and a dedicated AI server for intelligent chat functionality.
---------------------	---

Realisation	The frontend is built with React for a responsive, user-friendly interface. The .NET backend manages data storage and processing, while a Python/Ollama-based AI server handles LLM operations for the chat. Docker containerization ensures easy deployment and scalability.
-------------	---

Results	The result is a fully functional 'EduShelf' platform featuring user authentication, document upload and indexing, and an interactive AI chat. Users can manage their document libraries and obtain instant answers from their study materials, demonstrating the effective integration of modern web technologies and AI in education.
---------	--

Illustrative graph, photo (incl. explanation)



Participation in competitions
Awards

Accessibility of
final project thesis

HTL HollabrunnAnton
Ehrenfriedstraße 10
2020 Hollabrunn

Approval (Date / Signature)

Examiner/s

Head of Department / College